



哈爾濱工業大學 (深圳)
HARBIN INSTITUTE OF TECHNOLOGY

实验报告

开课学期: 2024 秋季
课程名称: 数据库系统
实验名称: 数据库系统功能实现
实验性质: 设计型
实验学时: 8 地点: T2109
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实验与创新实践教育中心制

2024 年 9 月

1 实验环境

- 操作系统: Linux
- 编程语言: C++
- 编译环境: g++
- 依赖库: Linux 标准库

2 测试结果

请截图测试结果

2.1 任务 1.1 磁盘存储管理器测试截图

disk_manager.cpp

```
In [11]: !make disk_manager_test

[ 20%] Built target gtest
[ 40%] Built target gtest_main
[ 80%] Built target storage
[ 90%] Linking CXX executable ../bin/disk_manager_test
[100%] Built target disk_manager_test

In [12]: ! ./bin/disk_manager_test

Running main() from /content/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 2 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 2 tests from DiskManagerTest
[ RUN    ] DiskManagerTest.FileOperation
[ OK     ] DiskManagerTest.FileOperation (3 ms)
[ RUN    ] DiskManagerTest.PageOperation
[ OK     ] DiskManagerTest.PageOperation (13 ms)
[-----] 2 tests from DiskManagerTest (16 ms total)

[-----] Global test environment tear-down
[=====] 2 tests from 1 test suite ran. (17 ms total)
[ PASSED ] 2 tests.
```

2.2 任务 1.2 缓冲池替换策略测试截图

lru_replacer

```
In [13]: !make lru_replacer_test

[ 25%] Built target gtest
[ 50%] Built target gtest_main
[ 75%] Built target lru_replacer
[100%] Built target lru_replacer_test

In [14]: ! ./bin/lru_replacer_test

Running main() from /content/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 3 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 3 tests from LRUCacheTest
[ RUN    ] LRUCacheTest.SimpleTest
[ OK     ] LRUCacheTest.SimpleTest (0 ms)
[ RUN    ] LRUCacheTest.MixTest
[ OK     ] LRUCacheTest.MixTest (19 ms)
[ RUN    ] LRUCacheTestConcurrencyTest
[ OK     ] LRUCacheTestConcurrencyTest (145 ms)
[-----] 3 tests from LRUCacheTest (164 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 1 test suite ran. (165 ms total)
[ PASSED ] 3 tests.
```

2.3 任务 1.3 缓冲池管理器测试截图

buffer_pool_manager

```
In [15]: !make buffer_pool_manager_test

[ 20%] Built target gtest
[ 40%] Built target gtest_main
[ 80%] Built target storage
[100%] Built target buffer_pool_manager_test

In [16]: ! ./bin/buffer_pool_manager_test

Running main() from /content/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 4 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 4 tests from BufferPoolManagerTest
[ RUN    ] BufferPoolManagerTest.SimpleTest
[ OK     ] BufferPoolManagerTest.SimpleTest (74 ms)
[ RUN    ] BufferPoolManagerTest.LargeScaleTest
[ OK     ] BufferPoolManagerTest.LargeScaleTest (240 ms)
[ RUN    ] BufferPoolManagerTest.MultipleFilesTest
[ OK     ] BufferPoolManagerTest.MultipleFilesTest (1570 ms)
[ RUN    ] BufferPoolManagerTestConcurrencyTest
[ OK     ] BufferPoolManagerTestConcurrencyTest (1757 ms)
[-----] 4 tests from BufferPoolManagerTest (3642 ms total)

[-----] Global test environment tear-down
[=====] 4 tests from 1 test suite ran. (3642 ms total)
[ PASSED ] 4 tests.
```

2.4 任务 2 记录管理器测试截图

record_manager

In [17]:

```
!make record_manager_test  
  
[ 8%] Built target gtest  
[ 16%] Built target gtest_main  
[ 33%] Built target storage  
[ 45%] Built target index  
[ 58%] Built target recovery  
[ 70%] Built target transaction  
[ 79%] Built target system  
[ 91%] Built target record  
[100%] Built target record_manager_test
```

In [18]:

```
! ./bin/record_manager_test  
  
Running main() from /content/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc  
[=====] Running 2 tests from 1 test suite.  
[=====] Global test environment set-up.  
[=====] 2 tests from RecordManagerTest  
[ RUN      ] RecordManagerTest.SimpleTest  
insert 451  
delete 284  
update 265  
[ OK      ] RecordManagerTest.SimpleTest (658 ms)  
[ RUN      ] RecordManagerTest.MultipleFilesTest  
[ OK      ] RecordManagerTest.MultipleFilesTest (232 ms)  
[=====] 2 tests from RecordManagerTest (891 ms total)  
  
[=====] Global test environment tear-down  
[=====] 2 tests from 1 test suite ran. (891 ms total)  
[ PASSED  ] 2 tests.
```